MSG No. 97.154 $I42'2'$ [Type III, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `42'2'`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,2,3,4,5,6,7,8]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[9,10,11,12,13,14,15,16]`

Table 2: Wyckoff site: `2b`, site symmetry: `42'2'`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[1,2,3,4,5,6,7,8]`
2	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[9,10,11,12,13,14,15,16]`

Table 3: Wyckoff site: `4c`, site symmetry: `22'2'.`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ 0\right]$	`[1,4,5,6]`
2	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[2,3,7,8]`
3	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[9,12,13,14]`
4	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[10,11,15,16]`

Table 4: Wyckoff site: `4d`, site symmetry: `2.2'2'`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ \frac{1}{4}\right]$	`[1,4,15,16]`
2	$\left[\frac{1}{2},\ 0,\ \frac{1}{4}\right]$	`[2,3,13,14]`
3	$\left[0,\ \frac{1}{2},\ \frac{3}{4}\right]$	`[5,6,10,11]`
4	$\left[\frac{1}{2},\ 0,\ \frac{3}{4}\right]$	`[7,8,9,12]`

Table 5: Wyckoff site: `4e`, site symmetry: `4..`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2,3,4]`
2	$[0,\ 0,\ -z]$	`[5,6,7,8]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	`[9,10,11,12]`
4	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	`[13,14,15,16]`

Table 6: Wyckoff site: `8f`, site symmetry: `2..`

No.	position	mapping
1	$\left[0,\ \frac{1}{2},\ z\right]$	`[1,4]`
2	$\left[\frac{1}{2},\ 0,\ z\right]$	`[2,3]`
3	$\left[0,\ \frac{1}{2},\ -z\right]$	`[5,6]`

continued ...

Table 6

No.	position	mapping
4	$\left[\frac{1}{2},\, 0,\, -z\right]$	[7,8]
5	$\left[\frac{1}{2},\, 0,\, z+\frac{1}{2}\right]$	[9,12]
6	$\left[0,\, \frac{1}{2},\, z+\frac{1}{2}\right]$	[10,11]
7	$\left[\frac{1}{2},\, 0,\, \frac{1}{2}-z\right]$	[13,14]
8	$\left[0,\, \frac{1}{2},\, \frac{1}{2}-z\right]$	[15,16]

Table 7: Wyckoff site: 8g, site symmetry: ..2'

No.	position	mapping
1	$[x,\, x,\, 0]$	[1,7]
2	$[-x,\, x,\, 0]$	[2,6]
3	$[x,\, -x,\, 0]$	[3,5]
4	$[-x,\, -x,\, 0]$	[4,8]
5	$\left[x+\frac{1}{2},\, x+\frac{1}{2},\, \frac{1}{2}\right]$	[9,15]
6	$\left[\frac{1}{2}-x,\, x+\frac{1}{2},\, \frac{1}{2}\right]$	[10,14]
7	$\left[x+\frac{1}{2},\, \frac{1}{2}-x,\, \frac{1}{2}\right]$	[11,13]
8	$\left[\frac{1}{2}-x,\, \frac{1}{2}-x,\, \frac{1}{2}\right]$	[12,16]

Table 8: Wyckoff site: 8h, site symmetry: .2'.

No.	position	mapping
1	$[x,\, 0,\, 0]$	[1,5]
2	$[0,\, x,\, 0]$	[2,7]
3	$[0,\, -x,\, 0]$	[3,8]
4	$[-x,\, 0,\, 0]$	[4,6]
5	$\left[x+\frac{1}{2},\, \frac{1}{2},\, \frac{1}{2}\right]$	[9,13]
6	$\left[\frac{1}{2},\, x+\frac{1}{2},\, \frac{1}{2}\right]$	[10,15]
7	$\left[\frac{1}{2},\, \frac{1}{2}-x,\, \frac{1}{2}\right]$	[11,16]
8	$\left[\frac{1}{2}-x,\, \frac{1}{2},\, \frac{1}{2}\right]$	[12,14]

Table 9: Wyckoff site: 8i, site symmetry: .2'.

No.	position	mapping
1	$\left[x,\, 0,\, \frac{1}{2}\right]$	[1,5]
2	$\left[0,\, x,\, \frac{1}{2}\right]$	[2,7]
3	$\left[0,\, -x,\, \frac{1}{2}\right]$	[3,8]
4	$\left[-x,\, 0,\, \frac{1}{2}\right]$	[4,6]
5	$\left[x+\frac{1}{2},\, \frac{1}{2},\, 0\right]$	[9,13]
6	$\left[\frac{1}{2},\, x+\frac{1}{2},\, 0\right]$	[10,15]
7	$\left[\frac{1}{2},\, \frac{1}{2}-x,\, 0\right]$	[11,16]
8	$\left[\frac{1}{2}-x,\, \frac{1}{2},\, 0\right]$	[12,14]

Table 10: Wyckoff site: 8j, site symmetry: ..2'

No.	position	mapping
1	$\left[x,\ x+\frac{1}{2},\ \frac{1}{4}\right]$	[1,15]
2	$\left[\frac{1}{2}-x,\ x,\ \frac{1}{4}\right]$	[2,14]
3	$\left[x+\frac{1}{2},\ -x,\ \frac{1}{4}\right]$	[3,13]
4	$\left[-x,\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	[4,16]
5	$\left[x,\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	[5,11]
6	$\left[-x,\ x+\frac{1}{2},\ \frac{3}{4}\right]$	[6,10]
7	$\left[x+\frac{1}{2},\ x,\ \frac{3}{4}\right]$	[7,9]
8	$\left[\frac{1}{2}-x,\ -x,\ \frac{3}{4}\right]$	[8,12]

Table 11: Wyckoff site: 16k, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$[-y,\ x,\ z]$	[2]
3	$[y,\ -x,\ z]$	[3]
4	$[-x,\ -y,\ z]$	[4]
5	$[x,\ -y,\ -z]$	[5]
6	$[-x,\ y,\ -z]$	[6]
7	$[y,\ x,\ -z]$	[7]
8	$[-y,\ -x,\ -z]$	[8]
9	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	[9]
10	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	[10]
11	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[11]
12	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$	[12]
13	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	[13]
14	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	[14]
15	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	[15]
16	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	[16]